

Flavor-Dependent Machine-Learned Jet Calibration

DFM-NOTE-2026-02

UTA Detector Foundation Model group

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August 2026 • github.com/afarbin/DetectorFoundationModel, branch dfm-merge**Abstract**

Extending the machine-learned jet p_T calibration of DFM-NOTE-2026-01 to heavy-flavor jets, we rebuild the per-jet dataset with all flavors (1.20M matched jets: $\sim 850k$ light, 150k c , 556k b , hadron-cone labels) and compare three calibration strategies on the best backbone of Note 01: a flavor-blind shared model, a flavor-conditioned shared model (truth-flavor one-hot into the head), and dedicated per-flavor models. The uncorrected response differs by flavor as expected (light +4.4% at 65–90 GeV; b on-scale with narrower core in $t\bar{t}$). Dedicated per-flavor models lose clearly to the shared model for c (JER 0.154 vs 0.131) and b (0.138 vs 0.128): cross-flavor statistics outweigh specialization at these sample sizes. Flavor conditioning is the best strategy overall (best closure, 3.9% RMS; improves b), but presumes flavor knowledge at inference—directly motivating the joint tagging–calibration architecture of DFM-NOTE-2026-03. The b -jet corrected response exhibits the expected semileptonic low-side shoulder rather than a resolved second mode.

1 Introduction

DFM-NOTE-2026-01 [1] established machine-learned jet p_T calibration on light jets, halving the resolution of the input reconstruction with a tracks + jet-features + cells model (TJC-graph). Jet response is, however, flavor-dependent: b and c jets differ in shower composition, and semileptonic heavy-flavor decays lose neutrino energy—a known pain point of classical calibration chains. This note answers a practical question: given finite samples, should heavy-flavor jets be calibrated by dedicated models, a shared model, or a shared model conditioned on flavor? All dataset machinery, cuts, metrics, and the TJC-graph backbone are as in Ref. [1], Sections 3–4; only the flavor selection changes.

2 Dataset

The builder retains all truth-matched jets with hadron-cone labels in $\{0, 4, 5\}$ (light/ c/b ; taus excluded), storing the label per jet: 1,197,615 jets over the 17 files—approximately 850k light, 150k c , 556k b matched jets before isolation ($t\bar{t}$ is b -rich by construction). Splits, quality cuts, isolation flags, and spectrum weights are identical to Ref. [1].

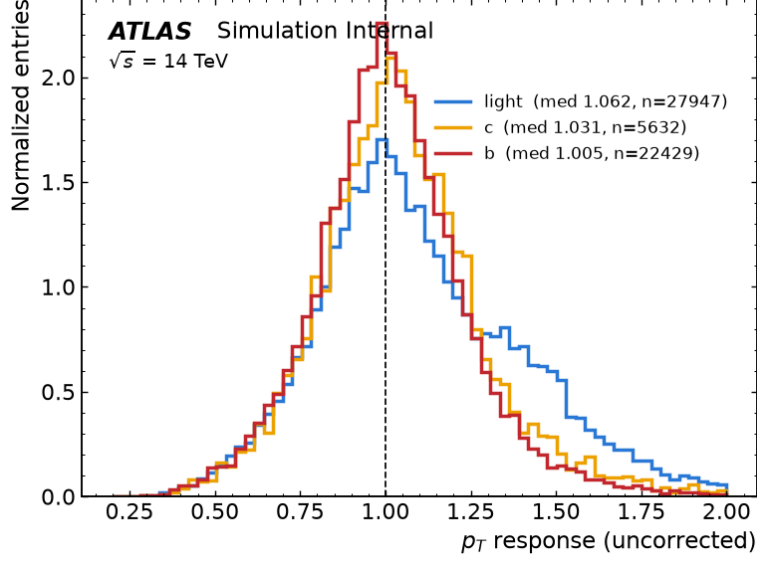


Figure 1: Uncorrected p_T response by flavor (isolated test jets): light jets read +4.4% high at 65–90 GeV; b jets are on-scale with a narrower core—in $t\bar{t}$ the semileptonic deficit is partly offset by b jets being harder and more central. The shape differences are what a flavor-aware calibration exploits.

evaluated on	uncorrected	blind shared	flavor-conditioned	dedicated
light JER@65–90	0.230	0.1152	0.1150	0.1152
c JER@65–90	0.182	0.1305	0.1328	0.1536
b JER@65–90	0.176	0.1305	0.1283	0.1384
overall closure RMS	—	4.3%	3.9%	—

Table 1: JER at 65–90 GeV per flavor under the three strategies (TJC-graph backbone, 3 seeds).

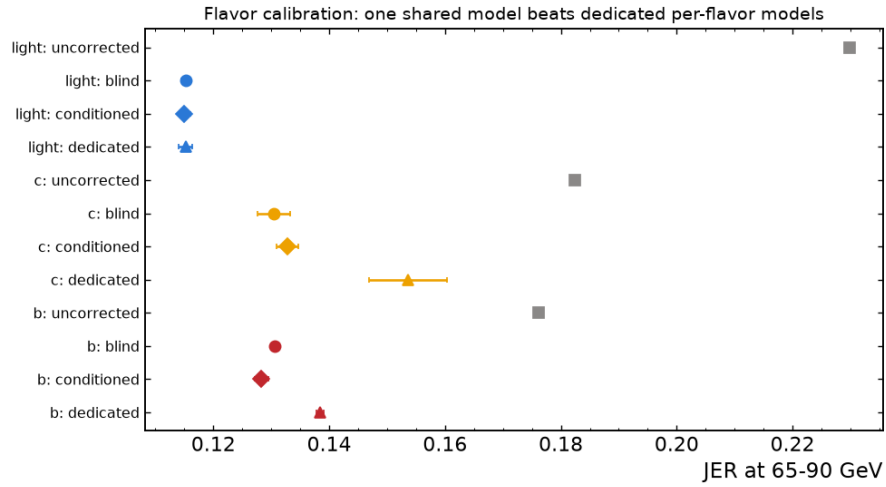


Figure 2: JER at 65–90 GeV, flavor $\times$ strategy: squares = uncorrected, circles = blind shared, diamonds = truth-flavor conditioned, triangles = dedicated. Dedicated models are visibly worse for c and b .

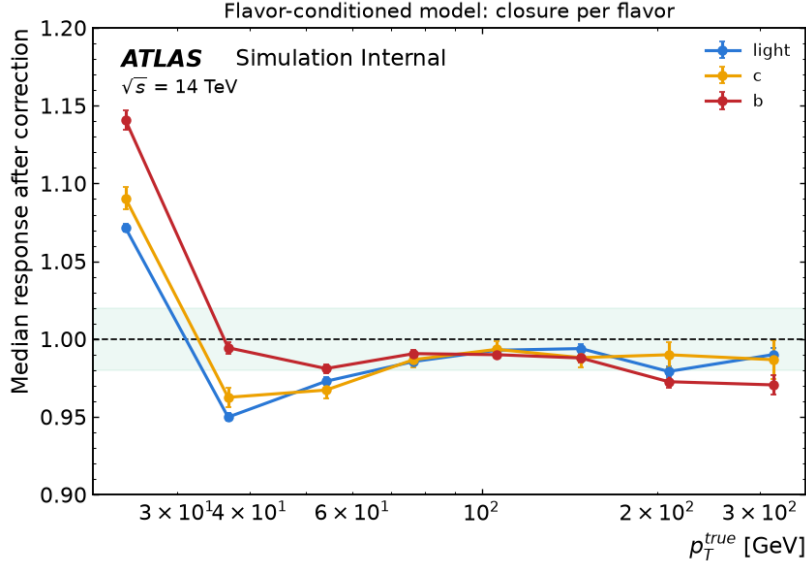


Figure 3: Flavor-conditioned model: closure per flavor within $\sim 2\%$ above 45 GeV; the shared low- p_T non-closure of Note 01 remains.

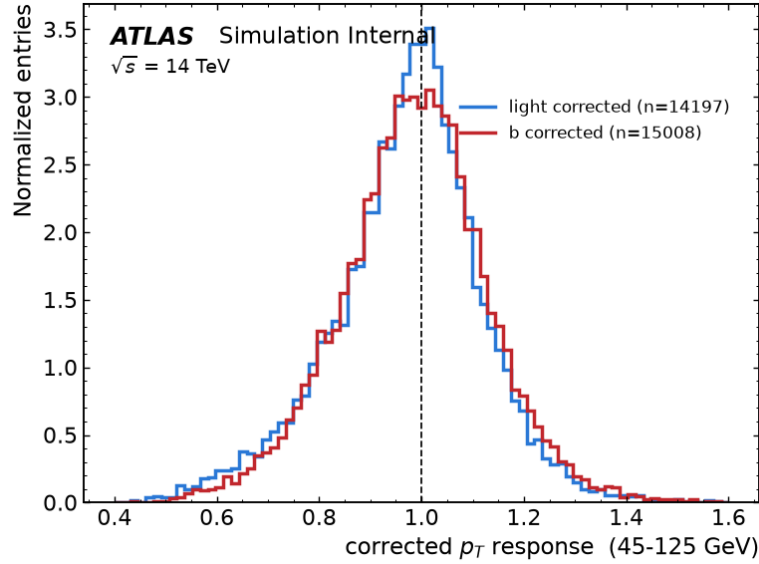


Figure 4: Corrected b -jet response versus light at 45–125 GeV: a broadened low-side shoulder from semileptonic decays rather than a resolved second mode; a single Gaussian per-jet σ remains defensible, with a mixture-density head as the identified upgrade path.

3 Strategies and results

4 Conclusions

One shared model beats separate calibrations at these statistics: dedicated c and b models are starved of data relative to what cross-flavor sharing provides. **Flavor conditioning is the best overall strategy**—matching the blind model for light and c while improving b and overall closure—but requires flavor knowledge at inference. The natural resolution is to predict flavor and calibration jointly, which is the subject of DFM-NOTE-2026-03 [2].

References

- [1] DFM-NOTE-2026-01, *Machine-Learned Jet p_T Calibration from Low-Level Detector Information*, [notes/DFM-NOTE-2026-01-jet-pt-calibration/](#).
- [2] DFM-NOTE-2026-03, *Panoptic Flavor Tagging and Jet Calibration*, [notes/DFM-NOTE-2026-03-panoptic-tagging-calibration/](#).